

UI/UX dan Source Code Framework Physiological Digital Twin CAPAR

1. Konsep UI/UX Digital Twin Dashboard

Antarmuka dirancang untuk menampilkan kondisi physiological digital twin secara longitudinal. UI tidak menampilkan hanya skor risiko, tetapi memperlihatkan state, trajectory, evidence, uncertainty, dan mekanisme recovery.

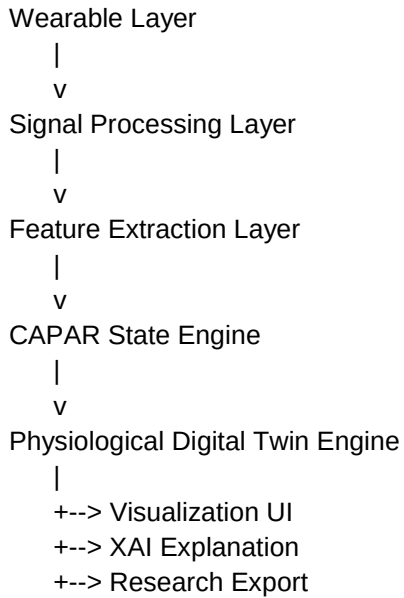
2. Struktur Menu Utama

Menu	Fungsi
Dashboard Resilience	Ringkasan lima domain: Vulnerability, Cardiac Reserve, Autonomic Reserve, Recovery Capacity, Regulation Stability.
Live Physiological State	Monitoring HR, RR, HRV, DFA $\alpha 1$, activity context, dan current CAPAR state.
Episode Explorer	Detail episode deviation, persistence, recovery, relapse.
Digital Twin State	Visualisasi latent physiological state dan perubahan antar waktu.
Explainable AI	Supporting evidence, contradicting evidence, uncertainty.
Model & Governance	Threshold, baseline, model version, data quality, provenance.
Export & Research	Dataset, laporan evaluasi, trajectory, dan eksperimen E1-E6.

3. Desain Dashboard Resilience

+-----+	
Physiological Digital Twin - CAPAR	
+-----+	
Resilience Score	
Vulnerability 72	
Cardiac Reserve 84	
Autonomic Reserve 78	
Recovery Capacity 90	
Regulation Stability 82	
+-----+	
Current State: Recovery	
Episode: Walking -> Recovery	
TTR: 3.2 min	
+-----+	
Evidence	
HR slope ↓ RMSSD ↑ DFA stable	
+-----+	

4. Arsitektur Software



5. Source Code Contoh - Data Model

```
class PhysiologicalTwinState:
    def __init__(self):
        self.vulnerability = 0
        self.cardiac_reserve = 0
        self.autonomic_reserve = 0
        self.recovery_capacity = 0
        self.regulation_stability = 0

    def resilience_score(self):
        return (
            0.2*(100-self.vulnerability)
            +0.2*self.cardiac_reserve
            +0.2*self.autonomic_reserve
            +0.2*self.recovery_capacity
            +0.2*self.regulation_stability
        )
```

6. Source Code - Feature Mapping CAPAR

```
def capar_to_latent(features):

    latent = {}

    latent['chronotropic_response'] = (
```

```

        features['HR_mean'] +
        features['HR_slope']
    )

    latent['vagal_control'] = (
        features['RMSSD'] +
        features['pNN50']
    )

    latent['autonomic_complexity'] = (
        features['DFA_alpha1']
    )

    latent['dynamic_stability'] = (
        1 / (1 + features['local_variance'])
    )

    latent['recovery_dynamics'] = {
        'TTR': features['TTR'],
        'slope': features['recovery_slope'],
        'relapse': features['relapse_count']
    }

    return latent

```

7. Source Code - Resilience Estimation

```

def estimate_resilience(latent):

    cardiac_reserve = normalize(
        latent['chronotropic_response']
    )

    autonomic_reserve = normalize(
        latent['vagal_control']
        + latent['autonomic_complexity']
    )

    recovery_capacity = normalize(
        latent['recovery_dynamics']['slope']
        - latent['recovery_dynamics']['TTR']
    )

    stability = normalize(
        latent['dynamic_stability']
        - latent['recovery_dynamics']['relapse']
    )

```

```

return {
  'cardiac_reserve': cardiac_reserve,
  'autonomic_reserve': autonomic_reserve,
  'recovery_capacity': recovery_capacity,
  'stability': stability
}

```

8. Visualisasi State Transition

Baseline Mature

|

v

No Deviation

|

v

Deviation

|

+--> Persistent Deviation

|

v

Recovery Start

|

v

Full Recovery

|

v

Baseline Update

Relapse:

Recovery Start --> Persistent Deviation

9. Komponen XAI

Setiap keputusan digital twin harus menyediakan alasan:

- Evidence pendukung: feature yang meningkatkan confidence.
- Evidence kontra: feature yang melemahkan interpretasi.
- Uncertainty: kualitas data, jumlah baseline, dan confidence model.
- Traceability: sumber data sampai keputusan.

10. Hubungan dengan Disertasi CAPAR

UI/X ini merepresentasikan kontribusi CAPAR sebagai physiological digital twin: mengubah wearable signal menjadi estimasi state fisiologis personal yang dapat dijelaskan, divalidasi, dan digunakan untuk analisis longitudinal.